








Advanced Time Series Analysis and Forecasting

Seminar 12: Machine learning and deep learning for time series



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Acronyms used in this material (1/2)

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **ARX**: AutoRegressive model with eXogenous variables
- **CV**: Cross-Validation
- **DM**: Diebold–Mariano (test)
- **ENTSO-E**: European Network of Transmission System Operators for Electricity
- **EU**: European Union
- **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)
- **GW**: Giacomini–White (test of conditional predictive ability)
- **HAR**: Heterogeneous AutoRegressive (model)
- **HICP**: Harmonised Index of Consumer Prices
- **HLN**: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- **HU**: Hungary (Eurostat country code)
- **LSTM**: Long Short-Term Memory (network)

Acronyms used in this material (2/2)

- **MAE**: Mean Absolute Error
- **MASE**: Mean Absolute Scaled Error
- **MCS**: Model Confidence Set
- **MLP**: MultiLayer Perceptron
- **MSE**: Mean Squared Error
- **MW**: MegaWatt
- **OLS**: Ordinary Least Squares
- **OOS**: Out-Of-Sample
- **OWA**: Overall Weighted Average (of sMAPE and MASE relative to Naive 2)
- **QLIKE**: Quasi-LIKElihood loss
- **QRF**: Quantile Regression Forest
- **RMSE**: Root Mean Squared Error
- **RNN**: Recurrent Neural Network
- **RV**: Realised Variance
- **SE**: Standard Error
- **SHAP**: SHapley Additive exPlanations
- **SPA**: Superior Predictive Ability (test)
- **TCN**: Temporal Convolutional Network

Today's question and seminar structure

- The question of the seminar
  - **how do we validate, fit and compare machine-learning and deep models on time series without fooling ourselves?**
- Outputs to produce
  - the answers to the A exercises
  - one figure and one table for each B exercise
  - the analysis requested in C1 and the verdicts on the AI answer in C2, each with one line of justification
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- The exercises, their types and their order: the exercise map (next slide)
- Seminar: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	lasso and ridge under an orthonormal design; adaptive lasso and elastic net	Solved, Proposed	A1
A3, A4	the variance of bagging; boosting as gradient descent	Solved, Proposed	A3
A5, A6	gradients through a scalar RNN; the LSTM forget gate	Solved, Proposed	A5
A7, A8	attention by hand; DLinear as one linear map	Solved, Proposed	A7
B1, B2	validation of HAR; a random forest with a time index	Solved, Proposed	B1
B3, B4	local and global AR for EU inflation; the panel lasso for Romania	Solved, Proposed	B3
B5, B6	QRF and ARX for the evening peak; monotone boosting	Solved, Proposed	B5
B7, B8	LSTM against HAR; DLinear and N-BEATS on 100 M4 series	Solved, Proposed	B7
C1, C2	deep learners for crypto volatility; what is wrong in an AI answer?	Proposed	B7, A1–A8

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in the seminar
- Part A: derivations and computations on paper; Part B: real data, each exercise ending with an interpretation question; Part C: open questions and the critique of an AI answer

Data used

Series	Source	Frequency	Period
RV of FTSE 100, DAX	Heber et al. (2009)	daily, 5-minute returns	2000–2022
HICP inflation, 27 EU countries	Eurostat	monthly	2000–2026
Romanian load	Energy-Charts (ENTSO-E)	hourly	2023–2026
M4 hourly series	Makridakis, Spiliotis and Assimakopoulos (2020)	hourly	–
RV of Bitcoin, Ether	Binance one-minute prices	daily	2018–2026

- RV in $\%^2$, load in GW (MW in the answers)
- Functions in the notebook
  - `cv_estimates`, `global_local`, `penalised_fit`, `QRF`, `rv_forecasts`, `m4_global`, `dm_test`, `mcs`
- No account or key is needed

Prerequisites for Today (1/24): the lag matrix and forecast errors

- **Lag matrix**: time series turned into a regression data set

$$\text{row } t: \quad \big(y_{t+h};\ y_t, y_{t-\mathbf{1}}, \ldots, y_{t-p+\mathbf{1}}\big), \qquad \hat{y}_{t+h} = f(y_t, \ldots, y_{t-p+\mathbf{1}})$$

  - y_{t+h}: target; p: number of lags (features); h: horizon; f: any learner (OLS, tree, network)
  - **direct** forecasts: one model per horizon h; **recursive**: a one-step model iterated h times

- **Error measures** over T forecasts, $e_t = y_t - \hat{y}_t$

$$\mathrm{RMSE} = \sqrt{\frac{1}{T}\sum_t e_t^{\mathbf{2}}}, \qquad \mathrm{MAE} = \frac{1}{T}\sum_t |e_t|$$

  - RMSE: the square root of the mean squared error (MSE); MAE: mean absolute error, less sensitive to outliers

- **Windows**
  - rolling: the last W observations; expanding: all observations up to the forecast origin; refit every k periods

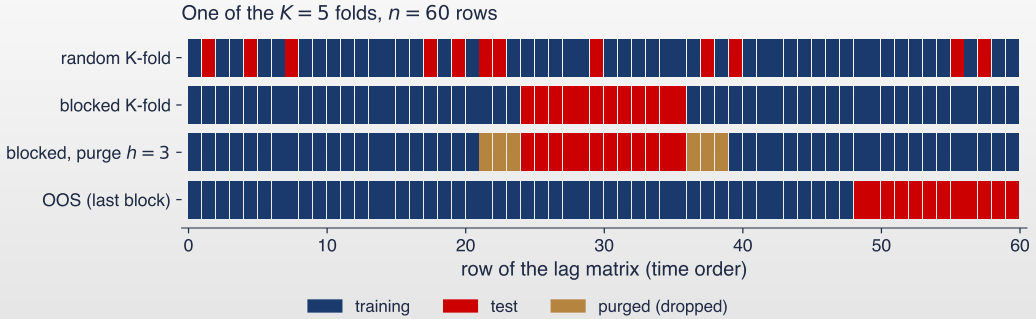
Prerequisites for Today (2/24): cross-validation schemes

- **Cross-validation** (CV): estimate the future error by fitting on part of the rows and testing on the rest

$$\widehat{\mathrm{MSE}}_{\mathrm{CV}} = \frac{1}{K}\sum_{k=1}^{K}\frac{1}{|F_k|}\sum_{t\in F_k}\left(y_{t+h} - \hat{f}^{(-k)}(x_t)\right)^2$$

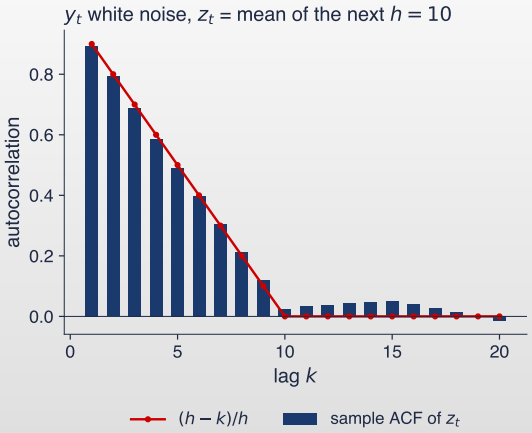
  - F_k: rows of fold k; $|F_k|$: their number; $\hat{f}^{(-k)}$: the learner fitted without fold k; x_t: the lags in row t
- Four ways of choosing the folds
  - **random K-fold**: rows assigned to the K folds at random
  - **blocked K-fold**: K contiguous blocks of rows
  - **purge**: drop h training rows on each side of the test block, so no training target overlaps a test target
  - **OOS** (out of sample): train on the first part, test only on the last block
- **Leakage**: information from the test rows reaches the training rows; the CV error is then too optimistic

Prerequisites for Today (3/24): the four schemes for one fold



- Random folds put test rows between training rows: their neighbours, almost the same numbers for persistent targets, are used for training
- Purging removes the h rows next to the test block; OOS uses only the past to predict the future, as in real forecasting

Prerequisites for Today (4/24): when random CV is valid



- Bergmeir, Hyndman and Koo (2018): random K-fold is valid for a well-specified autoregression
  - condition: **uncorrelated** errors u_{t+h}
  - it fails with a time-index feature or overlapping targets
- **Overlapping target**: the mean of the next h values

$$z_t = \frac{1}{h}\sum_{j=\mathbf{1}}^{h} y_{t+j}, \qquad \operatorname{corr}(z_t, z_{t+k}) = \frac{h-k}{h}$$

  - equality for white noise y, $k < h$; positively autocorrelated y: higher
  - neighbouring targets share $h-1$ terms
- Chart: $h = 10$, the sample ACF follows the line $(h-k)/h$

Prerequisites for Today (5/24): local, global and HAR models

- N series $y_{i,t}$, $i = 1, \ldots, N$ Montero-Manso and Hyndman (2021)
  - **local** model: one per series, $y_{i,t+h} = f_i(x_{i,t})$, about T rows for each
  - **global** model: the same f for all series, $y_{i,t+h} = f(x_{i,t})$, fitted on NT stacked rows (pooled OLS for an AR)
  - global: more rows per coefficient, less variance; the price is a bias if the series differ
- **HAR** Corsi (2009) for $y_t = \log RV_t$

$$y_{t+h} = \beta_{\mathbf{0}} + \beta_d y_t + \beta_w \bar{y}_t^{(\mathbf{5})} + \beta_m \bar{y}_t^{(\mathbf{22})} + u_{t+h}, \qquad \bar{y}_t^{(K)} = \frac{1}{K} \sum_{j=\mathbf{0}}^{K-\mathbf{1}} y_{t-j}$$

  - RV_t: realised variance, the sum of squared intraday returns; daily, weekly and monthly means; OLS, one regression per horizon
  - lognormal correction: $\widehat{RV}_{t+h} = \exp(\hat{y}_{t+h} + \hat{s}^{\mathbf{2}}/2)$, $\hat{s}^{\mathbf{2}}$: residual variance

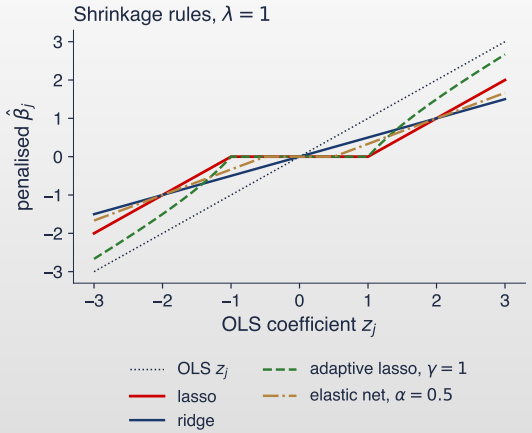
Prerequisites for Today (6/24): penalised regression

- Least squares plus a penalty on the coefficients (predictors standardised, intercept not penalised)

$$\hat{\beta} = \arg\min_{\beta} \frac{1}{2n}\|y - X\beta\|^2 + \lambda P(\beta)$$

  - n: rows; X: $n \times p$ predictors; $\lambda \geq 0$: penalty, chosen by CV; $\lambda = 0$: OLS
- Penalties $P(\beta)$
  - **lasso** Tibshirani (1996): $\sum_j |\beta_j|$: sets some coefficients exactly to zero (selection)
  - **ridge**: $\frac{1}{2}\sum_j \beta_j^2$: shrinks all coefficients, never to zero
  - **elastic net** Zou and Hastie (2005): $\alpha \sum_j |\beta_j| + \frac{1-\alpha}{2}\sum_j \beta_j^2$, $\alpha \in [0, 1]$
  - **adaptive lasso** Zou (2006): $\sum_j w_j|\beta_j|$, $w_j = 1/|\tilde{\beta}_j|^\gamma$ from a first estimate $\tilde{\beta}$, $\gamma > 0$
- **Post-lasso OLS**: OLS on the predictors selected by the lasso (removes the shrinkage bias)

Prerequisites for Today (7/24): shrinkage coefficient by coefficient



- Orthonormal predictors ($X'X = nI$), OLS coefficient z_j: each penalty acts separately

$$S(z, \lambda) = \text{sign}(z)(|z| - \lambda)_+$$

  - $(u)_+ = \max(u, 0)$; S: soft thresholding
- The rules in the chart
  - lasso $S(z_j, \lambda)$; ridge $z_j/(1 + \lambda)$
  - adaptive lasso: threshold $\lambda/|z_j|^\gamma$
  - elastic net $S(z_j, \lambda\alpha)/(1 + \lambda(1 - \alpha))$
- Reading: the vertical gap to the dotted OLS line is the shrinkage bias

Prerequisites for Today (8/24): trees and random forests

- **Regression tree**: a piecewise constant function on a partition of the predictor space

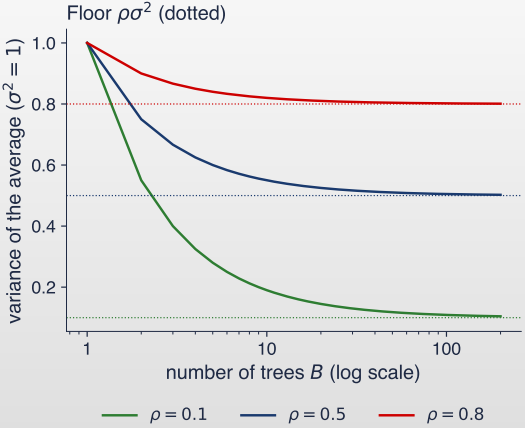
$$T(x) = \sum_{m=\mathbf{1}}^{M} c_m \mathbf{1}\{x \in R_m\}, \qquad c_m = \text{mean of } y_i \text{ in } R_m$$

  - R_m: leaves (rectangles); $\mathbf{1}\{\cdot\}$: indicator; each split $x_j \le s$ is chosen to minimise the sum of squared errors
  - deep trees: low bias, high variance; leaf size and depth control the complexity
- **Bagging** Breiman (1996) and **random forest** Breiman (2001): average B trees
  - bagging: each tree on a bootstrap sample (rows drawn with replacement)
  - random forest: also a random subset of the predictors at each split, which decorrelates the trees
- **Quantile regression forest (QRF)** Meinshausen (2006): the whole conditional distribution

$$\hat{F}(y \mid x) = \sum_i w_i(x)\, \mathbf{1}\{y_i \le y\}, \qquad w_i(x) = \frac{1}{B} \sum_b \frac{\mathbf{1}\{x_i \in R_b(x)\}}{|R_b(x)|}$$

  - $R_b(x)$: the leaf of tree b that contains x; quantile $\hat{q}_\tau(x) = \inf\{y : \hat{F}(y \mid x) \ge \tau\}$

Prerequisites for Today (9/24): the variance of an average



- B predictors $\hat{f}_b$, each with variance σ^2 and pairwise correlation ρ

$$\mathrm{Var}\Big(\frac{1}{B}\sum_b \hat{f}_b\Big) = \rho\sigma^2 + \frac{(1-\rho)\sigma^2}{B}$$

  - the second term vanishes as B grows
  - the first one does not: a floor $\rho\sigma^2$
- Chart: the curves reach their floor after a few dozen trees
- Lower ρ (decorrelated trees) lowers the floor: the idea of the random forest

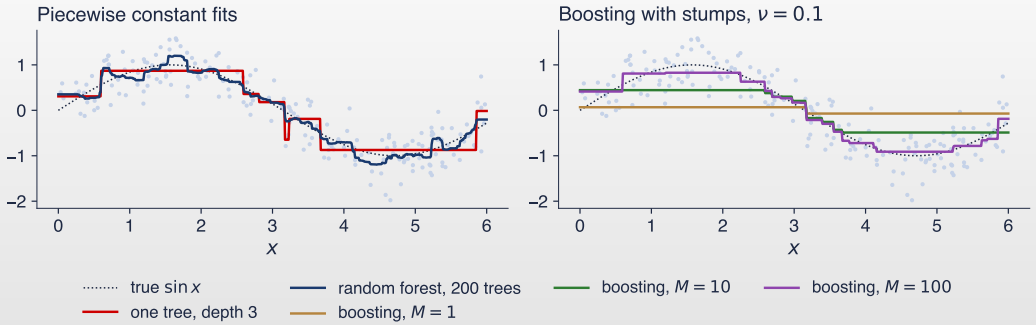
Prerequisites for Today (10/24): gradient boosting

- **Gradient boosting** Friedman (2001): add small trees fitted to the negative gradient of the loss

$$F_{\mathbf{0}} = \text{const}, \qquad g_{i,m} = -\frac{\partial \ell(y_i, F)}{\partial F}\Big|_{F=F_{m-\mathbf{1}}(x_i)}, \qquad F_m = F_{m-\mathbf{1}} + \nu h_m$$

  - ℓ: loss; h_m: a small tree fitted to $(x_i, g_{i,m})$; $\nu \in (0, 1]$: learning rate (shrinkage); M: number of iterations
  - squared loss $\frac{\mathbf{1}}{\mathbf{2}}(y - F)^{\mathbf{2}}$: $g = y - F$, the residuals
  - small ν: more iterations, a more stable fit; M and ν are tuned together
- **Histogram gradient boosting**: predictors binned (e.g. 255 bins) before searching splits: fast on large data
- **Monotone constraint** in x_j: every split on x_j must keep the fit non-decreasing in x_j
  - partial dependence: $\mathrm{PD}_j(v) = \frac{\mathbf{1}}{n}\sum_i \hat{F}(v, x_{i,-j})$, the average prediction when x_j is set to v ($x_{i,-j}$: the other predictors of row i)

Prerequisites for Today (11/24): a tree, a forest and boosting



- Simulated $y = \sin x +$ noise, 200 points; left: one tree of depth 3 (8 leaves) and a forest of 200 trees, smoother because it averages
- Right: boosting with stumps (trees with one split) and $\nu = 0.1$ after 1, 10 and 100 iterations: the fit is built up step by step

Prerequisites for Today (12/24): neural networks

- **Multilayer perceptron** (MLP): layers of affine maps followed by a nonlinearity

$$a^{(l)} = \phi\big(W^{(l)} a^{(l-\mathbf{1})} + b^{(l)}\big), \quad a^{(\mathbf{0})} = x, \qquad \hat{y} = W^{(L)} a^{(L-\mathbf{1})} + b^{(L)}$$

  - $W^{(l)}$, $b^{(l)}$: weights and biases of layer l; ϕ: activation applied elementwise
  - $\tanh(u) \in (-1, 1)$; sigmoid $\sigma(u) = 1/(1 + e^{-u}) \in (0, 1)$; ReLU $\max(u, 0)$
- **Training**: minimise the average loss by stochastic gradient descent
  - $\theta \leftarrow \theta - \eta \nabla_\theta \frac{\mathbf{1}}{|B|} \sum_{i \in B} \ell(y_i, \hat{y}_i)$; B: mini-batch; η: learning rate; Adam: an adaptive variant
  - gradients by the chain rule (backpropagation); epoch: one pass over the training rows
  - seed: the random number that fixes initial weights and batches; results vary with it, so average several seeds
  - early stopping: stop when the loss on a validation block stops falling

Prerequisites for Today (13/24): recurrent networks and the LSTM

- **RNN**: a hidden state updated at every step; the gradient to a past state is a product of Jacobians

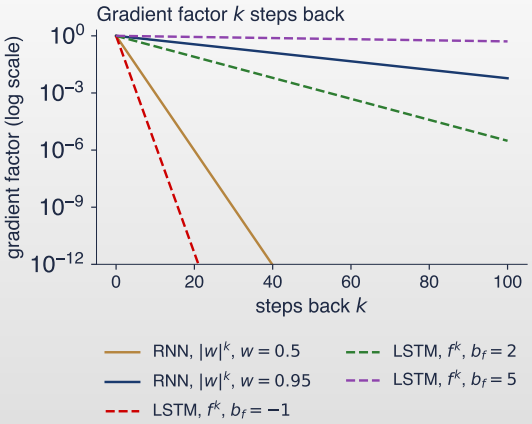
$$h_t = \tanh(W h_{t-\mathbf{1}} + U x_t + b), \qquad \frac{\partial h_T}{\partial h_t} = \prod_{k=t+\mathbf{1}}^{T} \operatorname{diag}(\mathbf{1} - h_k^{\mathbf{2}})\, W$$

  - h_t: hidden state; x_t: input; W, U, b: weights; norm $\|W\| < 1$: vanishing gradients, > 1: possibly exploding Bengio, Simard and Frasconi (1994)
- **LSTM** Hochreiter and Schmidhuber (1997): an additive memory cell c_t controlled by three gates in $(0, 1)$

$$f_t = \sigma(W_f x_t + U_f h_{t-\mathbf{1}} + b_f), \qquad i_t = \sigma(W_i x_t + U_i h_{t-\mathbf{1}} + b_i), \qquad o_t = \sigma(W_o x_t + U_o h_{t-\mathbf{1}} + b_o),$$
$$\tilde{c}_t = \tanh(W_c x_t + U_c h_{t-\mathbf{1}} + b_c), \qquad c_t = f_t \odot c_{t-\mathbf{1}} + i_t \odot \tilde{c}_t, \qquad h_t = o_t \odot \tanh(c_t)$$

  - f_t: forget gate; i_t: input gate; o_t: output gate; $\tilde{c}_t$: candidate memory; $\odot$: elementwise product
  - along the cell $\partial c_t / \partial c_{t-\mathbf{1}} = \operatorname{diag}(f_t)$, with no tanh in between; b_f: forget bias

Prerequisites for Today (14/24): vanishing gradients



- Scalar RNN: each step multiplies the gradient by $w(1 - h_t^2)$
  - so it is at most $|w|^k$ after k steps
- LSTM cell path with a constant forget gate $f = \sigma(b_f)$: factor f^k
  - $b_f = -1$: $f \approx 0.27$, gone in a few steps
  - $b_f = 5$: $f \approx 0.993$, a signal survives 100 steps
- An open forget gate (large b_f) keeps long memory Hochreiter and Schmidhuber (1997)

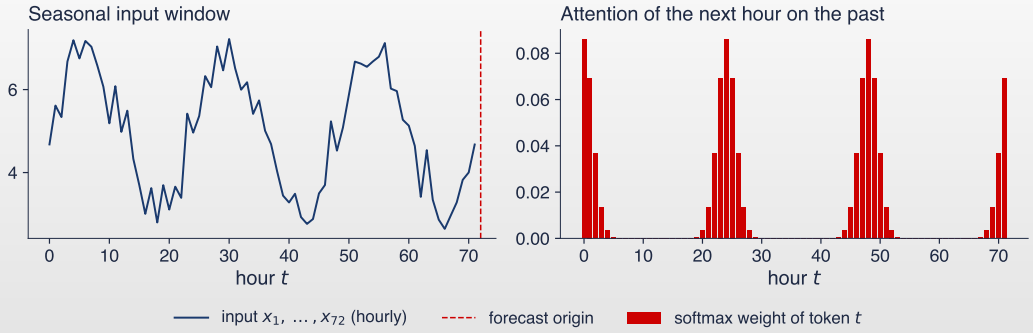
Prerequisites for Today (15/24): attention

- **Attention** Vaswani et al. (2017): each token takes a weighted average of the values of all tokens

$$\mathrm{Attn}(q, K, V) = \sum_{j=1}^{n} \alpha_j v_j, \qquad \alpha_j = \frac{\exp(q \cdot k_j / \sqrt{d_k})}{\sum_l \exp(q \cdot k_l / \sqrt{d_k})}$$

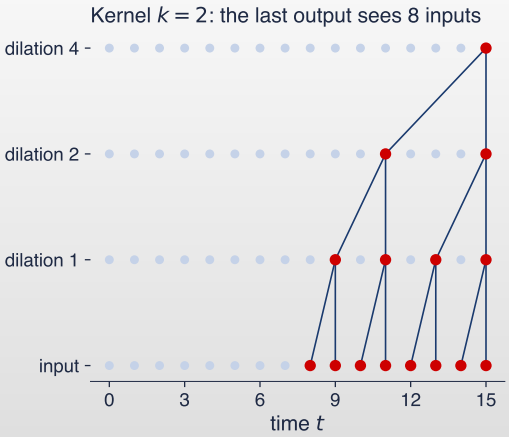
  - token: one input element (a time step or a patch); $q = W_Q x$: its query; $k_j = W_K x_j$: keys; $v_j = W_V x_j$: values
  - d_k: key dimension; softmax: positive weights α_j that sum to 1; similar key and query: large weight
- Order and position
  - the formula treats the tokens as a set; the order enters only through a positional encoding added to x_j (e.g. sines and cosines of the position)
  - patch Transformers (PatchTST Nie et al. (2023)): tokens are segments of 16–64 consecutive values, not single points
- **Occlusion**: the change in the forecast when one input (or patch) is masked; a direct measure of dependence

Prerequisites for Today (16/24): attention on a seasonal series



- Toy example: keys and query encode only the hour of the day; the query is the next hour (hour 72)
- The weights concentrate on the same hour of earlier days (0, 24, 48 hold 26% of the weight): attention finds seasonal analogues

Prerequisites for Today (17/24): temporal convolutional networks



- **Dilated causal convolution** Bai, Kolter and Koltun (2018)

$$y_t = \sum_{i=0}^{k-1} w_i x_{t-d\,i}$$

  - k: kernel size; d: dilation; causal: only x_t and the past
- **Receptive field**: how many past inputs one output sees
  - one convolution per level, dilations $1, 2, \ldots, 2^{\ell-1}$: $1 + (k-1)(2^\ell - 1)$
  - chart: $k = 2$, $\ell = 3$: 8 inputs
  - TCN blocks with two convolutions per level: $1 + 2(k-1)(2^\ell - 1)$

Prerequisites for Today (18/24): DLinear and N-BEATS

- **DLinear** Zeng et al. (2023): split the input window into trend and remainder, one linear map for each

$$\hat{y} = W_s(x - Ax) + W_t Ax$$

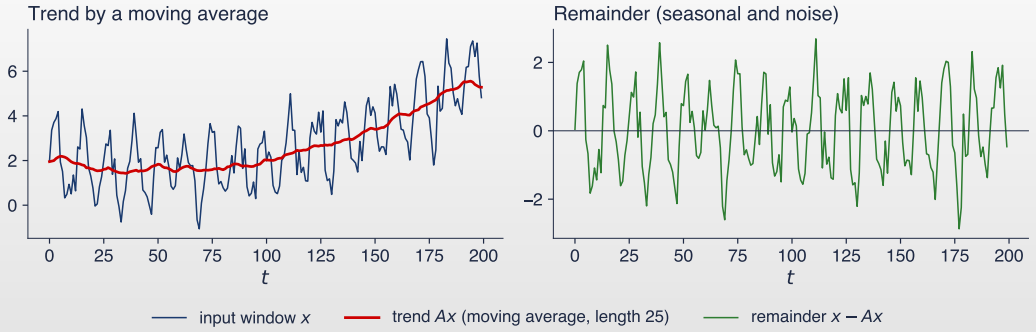
  - x: input window of length L; A: $L \times L$ moving-average matrix (trend Ax); W_s, W_t: $H \times L$ weights; H: horizon

- **N-BEATS** Oreshkin et al. (2020): a stack of fully connected blocks; block b removes what it explains from its input

$$x^{(b+1)} = x^{(b)} - \hat{x}^{(b)}, \qquad \hat{y} = \sum_b \hat{y}^{(b)}$$

  - $\hat{x}^{(b)}$: backcast (reconstruction of the input); $\hat{y}^{(b)}$: the block's partial forecast; trained as a global model on many series
  - ensemble: average of several models (seeds, input lengths, losses)

Prerequisites for Today (19/24): the DLinear decomposition



- Simulated window: quadratic trend, a 12-period season and noise; the moving average (length 25, ends repeated) gives Ax
- The remainder $x - Ax$ keeps the season and the noise; each part gets its own linear forecast

Prerequisites for Today (20/24): losses for variance and quantile forecasts

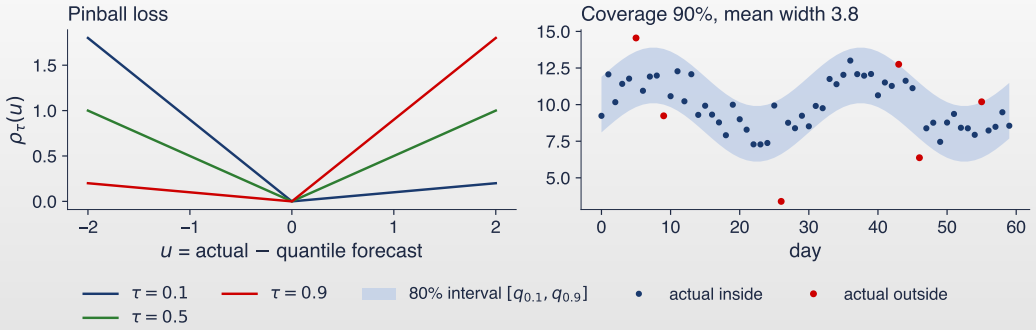
- **QLIKE** Patton (2011) for a variance forecast F of RV, and the **pinball loss** for a quantile

$$\text{QLIKE} = \frac{RV}{F} - \log \frac{RV}{F} - 1 \geq 0, \qquad \rho_\tau(u) = \max\left(\tau u, (\tau - 1)u\right)$$

  - QLIKE: zero only for $F = RV$; penalises under-prediction of the variance more
  - $u = y - \hat{q}_\tau$: actual minus quantile forecast; τ: quantile level; minimised in expectation by the true quantile
  - negative gradient in $F = \hat{q}_\tau$: $\tau - 1\{y < F\}$

- **Interval forecasts** from the deciles $\hat{q}_{\mathbf{0.1}}$, $\hat{q}_{\mathbf{0.9}}$
  - coverage: the share of actuals inside $[\hat{q}_{\mathbf{0.1}}, \hat{q}_{\mathbf{0.9}}]$, nominal 80%
  - width: the mean of $\hat{q}_{\mathbf{0.9}} - \hat{q}_{\mathbf{0.1}}$; good intervals are sharp (narrow) subject to correct coverage
  - average pinball over the deciles: a single score that rewards both

Prerequisites for Today (21/24): pinball loss and coverage



- Left: $\tau = 0.9$ charges 9 times more for an actual above the quantile than below it, so the forecast is pushed up
- Right: simulated 80% intervals cover 90% of the actuals: too wide (over-coverage)

Prerequisites for Today (22/24): the M4 measures

- M4 competition Makridakis, Spiliotis and Assimakopoulos (2020): horizon H, y_{n+h} actuals, $\hat{y}_{n+h}$ forecasts

$$\mathrm{sMAPE} = \frac{200}{H} \sum_{h=1}^{H} \frac{|y_{n+h} - \hat{y}_{n+h}|}{|y_{n+h}| + |\hat{y}_{n+h}|}, \qquad \mathrm{MASE} = \frac{\frac{1}{H} \sum_h |y_{n+h} - \hat{y}_{n+h}|}{\frac{1}{n-m} \sum_{t=m+1}^{n} |y_t - y_{t-m}|}$$

  - sMAPE: symmetric mean absolute percentage error, in %; MASE Hyndman and Koehler (2006): MAE scaled by the in-sample MAE of the seasonal naive
  - n: length of the training part; m: season length (24 for hourly data)

- Benchmarks and the overall score

$$\mathrm{OWA} = \frac{1}{2}\left(\frac{\mathrm{sMAPE}}{\mathrm{sMAPE}_{\text{Naive 2}}} + \frac{\mathrm{MASE}}{\mathrm{MASE}_{\text{Naive 2}}}\right)$$

  - seasonal naive: $\hat{y}_{n+h} = y_{n+h-m}$ (the same hour of the last day)
  - Naive 2: the last value of the seasonally adjusted series, reseasonalised; OWA < 1: better than Naive 2

Prerequisites for Today (23/24): comparing forecasts

- **Diebold–Mariano test** Diebold and Mariano (1995) (Chapter 1): d_t = loss of model 1 minus loss of model 2, $t = 1, \ldots, T$

$$DM = \frac{\bar{d}}{\sqrt{\hat{\Omega}/T}}, \qquad \hat{\Omega} = \hat{\gamma}_{\mathbf{0}} + 2\sum_{k=\mathbf{1}}^{h-\mathbf{1}} \hat{\gamma}_k, \qquad DM_{\mathrm{HLN}} = DM\sqrt{\frac{T + \mathbf{1} - 2h + h(h-1)/T}{T}}$$

  - $\bar{d}$: mean loss difference; $\hat{\gamma}_k$: autocovariances of d_t ($h - 1$ lags for h-step forecasts)
  - HLN: the Harvey–Leybourne–Newbold small-sample correction, compared with Student $t_{T-\mathbf{1}}$; $DM < 0$: model 1 better
- Many models at once
  - **MCS** Hansen, Lunde and Nason (2011): the model confidence set, the models not significantly worse than the best (by bootstrap)
  - **SPA** Hansen (2005): tests whether any of several models beats a benchmark, correcting for data snooping
- **Expert ARX for load** (Chapter 1) Ziel and Weron (2018): one OLS per hour s of day $d + 1$

$$L_{d+\mathbf{1},s} = \beta_{\mathbf{1}} L_{d,s} + \beta_{\mathbf{2}} L_{d-\mathbf{1},s} + \beta_{\mathbf{3}} L_{d-\mathbf{6},s} + \beta_{\mathbf{4}} L_d^{\min} + \beta_{\mathbf{5}} L_d^{\max} + \beta_{\mathbf{6}} L_{d,\mathbf{24}} + \sum_{j=\mathbf{1}}^{\mathbf{7}} \delta_j D_{j,d+\mathbf{1}} + \varepsilon$$

  - $L_{d,s}$: load of day d, hour s; $L_d^{\min}$, $L_d^{\max}$: minimum and maximum of day d; D_j: weekday dummies; quantiles from the empirical errors

Prerequisites for Today (24/24): Shapley values

- **Shapley value** Shapley (1953) of feature j for one prediction $\hat{f}(x)$, p features, $N = \{1, \dots, p\}$

$$\varphi_j = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!\,(p - |S| - 1)!}{p!} \left[v(S \cup \{j\}) - v(S) \right]$$

  - $v(S)$: the prediction when only the features in S are known; the bracket: the marginal contribution of j to coalition S
  - the weights average this contribution over all orders in which the features can be added

- **Interventional** SHAP Lundberg and Lee (2017): unknown features replaced by rows of a background sample

$$v(S) = \frac{1}{|\mathcal{B}|} \sum_{z \in \mathcal{B}} \hat{f}(x_S, z_{N \setminus S})$$

  - $\mathcal{B}$: background sample; x_S: the features of x in S; $z_{N \setminus S}$: the other features taken from z
  - exact computation needs 2^p coalitions; groups of features (e.g. all lags of one series) reduce p

A1: lasso and ridge under an orthonormal design [Solved]

Task

- OLS coefficients $z = (2.0, -0.6, 0.3, 1.2)$, $X'X = nI$, $\lambda = 0.5$.
- 1. Show that the lasso problem separates into $\frac{1}{2}(\beta_j - z_j)^2 + \lambda|\beta_j|$ and derive the soft-thresholding solution.
- 2. Compute the lasso and the ridge coefficients.
- 3. Which coefficients does each method set to zero?
- Report: one derivation; eight numbers; one sentence.

Solution

- 1. $\|y - X\beta\|^2/(2n) = \text{const} + \frac{1}{2}\|\beta - z\|^2$; subgradient $0 \in \beta_j - z_j + \lambda\partial|\beta_j|$ gives $\beta_j = 0$ if $|z_j| \le \lambda$, else $z_j - \lambda\,\text{sign}(z_j)$
- 2. Lasso: 1.50, −0.10, 0.00, 0.70; ridge: 1.333, −0.400, 0.200, 0.800
- 3. Lasso zeroes the third coefficient; ridge zeroes none
- Interpretation: lasso subtracts the same λ from every coefficient, so large coefficients are biased by λ; ridge shrinks proportionally and never selects

Back Exercise list

A2: adaptive lasso and elastic net [Proposed]

Solution discussed in class

Task

- The same z and $\lambda = 0.5$. Model: A1.
- 1. Adaptive lasso with weights $w_j = 1/|z_j|$ ($\gamma = 1$): compute the four thresholds and the coefficients.
- 2. Elastic net with $\alpha = 0.5$: compute the coefficients with the elastic-net rule.
- 3. Compare the bias on the largest coefficient across lasso, adaptive lasso and elastic net.
- Report: eight numbers; four numbers; one sentence.

A3: the variance of bagging [Solved]

Task

- B trees, each with variance $\sigma^2 = 1$, pairwise correlation $\rho = 0.3$.
- 1. Derive $\mathrm{Var}\left(\frac{1}{B}\sum_b \hat{f}_b\right) = \rho\sigma^2 + (1-\rho)\sigma^2/B$.
- 2. Compute it for $B = 1, 10, 100$ and its limit.
- 3. Which B brings the variance within 5% of its limit?
- Report: one derivation; four numbers; one number.

Solution

- 1. $\frac{1}{B^2}\left[\sum_b \mathrm{Var}\hat{f}_b + \sum_{b\neq c}\mathrm{Cov}(\hat{f}_b, \hat{f}_c)\right] = \frac{1}{B^2}\left[B\sigma^2 + B(B-1)\rho\sigma^2\right]$
- 2. 1.000, 0.370, 0.307; limit 0.30
- 3. $(1-\rho)/B \leq 0.05\rho \Rightarrow B \geq 47$
- Interpretation: beyond a few dozen trees only decorrelation (feature subsampling in random forests) lowers the variance

Back Exercise list

A4: boosting as gradient descent [Proposed]

Solution discussed in class

Task

- Squared loss $\ell = \frac{1}{2}(y - F)^2$, shrinkage $\nu = 0.1$, base learner: a constant (a tree without splits), $F_0 = 0$. Model: A3.
- 1. Show that the negative gradient is the residual and that $F_M = (1 - (1 - \nu)^M)\bar{y}$.
- 2. Compute $F_M/\bar{y}$ for $M = 10$ and 50, and the M needed to reach 90%.
- 3. Write the negative gradient of the pinball loss $\rho_\tau(y - F)$.
- Report: one derivation; three numbers; one formula.

A5: gradients through a scalar RNN [Solved]

Task

- $h_t = \tanh(wh_{t-\mathbf{1}} + ux_t)$ with $w = 0.9$.
- 1. Show that $\partial h_T / \partial h_{T-k} = \prod_{j=\mathbf{0}}^{k-\mathbf{1}} w(1 - h_{T-j}^{\mathbf{2}})$ and that $|\partial h_T / \partial h_{T-k}| \leq |w|^k$.
- 2. Compute the bound for $k = 10$ and $k = 50$.
- 3. If $|h_t| = 0.5$ along the path, compute the actual factor per step and the gradient at $k = 10$ and 50.
- Report: one derivation; two numbers; three numbers.

Solution

- 1. Chain rule: $\partial h_t / \partial h_{t-\mathbf{1}} = w \tanh'(\cdot) = w(1 - h_t^{\mathbf{2}})$, and $0 < 1 - h_t^{\mathbf{2}} \leq 1$
- 2. $0.9^{\mathbf{10}} = 0.349$, $0.9^{\mathbf{50}} = 0.00515$
- 3. Factor 0.675; 0.0196 at $k = 10$, $2.9 \times 10^{-\mathbf{9}}$ at $k = 50$
- Interpretation: saturation of the units speeds up the decay; a signal 50 steps back cannot move the weights

Back Exercise list

A6: the LSTM forget gate [Proposed]

Solution discussed in class

Task

- LSTM cell path $c_t = f_t c_{t-\mathbf{1}} + i_t \tilde{c}_t$, with a constant forget gate $f = \sigma(b_f)$. Model: A5.
- 1. Show that the direct path gives $\partial c_T / \partial c_{T-k} = f^k$.
- 2. Compute f for $b_f = 0, 1, 3$ and $f^{\mathbf{50}}$ in each case.
- 3. Why does the default initialisation not protect against vanishing gradients?
- Report: one derivation; six numbers; one sentence.

Back Exercise list

A7: attention by hand [Solved]

Task

- Keys $k_1 = (1, 0)$, $k_2 = (0, 1)$, $k_3 = (1, 1)$; values $v = (1, 2, 3)$; query $q = (1, 1)$; $d_k = 2$.
- 1. Compute the scores $qk_j/\sqrt{d_k}$.
- 2. Compute the softmax weights and the output.
- 3. What happens to the weights if the three tokens are permuted?
- Report: three numbers; four numbers; one sentence.

Solution

- 1. 0.707, 0.707, 1.414
- 2. Weights 0.248, 0.248, 0.503; output 2.255
- 3. The weights are permuted with the tokens and the output is unchanged: attention ignores order unless positions are encoded
- Interpretation: the critique of Zeng et al. starts here: a time series is an ordered object

Back Exercise list

A8: DLinear is one linear map [Proposed]

Solution discussed in class

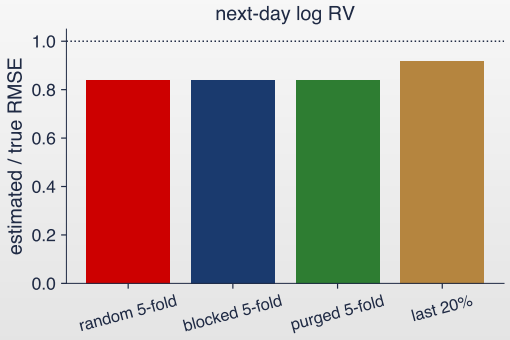
Task

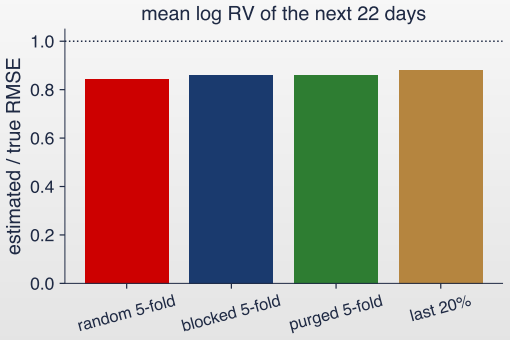
- $L = 3$ inputs, one output, moving average of length 3 with replicate padding; $W_s = (0.2, 0.3, 0.5)$, $W_t = (0.1, 0.1, 0.8)$. Model: A7.
- 1. Write the 3×3 matrix A with trend $= Ax$.
- 2. Show that the forecast is Wx with $W = W_s(I - A) + W_t A$ and compute W.
- 3. What does the result imply for the expressiveness of DLinear?
- Report: a matrix; three numbers; one sentence.

B1: validating HAR on FTSE 100 realised variance [Solved]

- **Question**: do random and blocked cross-validation estimate the future error of HAR, for one-day and 22-day targets?
- **Inputs**: Oxford-Man 5-minute RV of the FTSE 100, $n = 5\,543$ rows; in-set: the first 80% (to 5 September 2017); HAR on $\log RV$
- **Tasks**
  1. Estimate the RMSE on the in-set by random 5-fold, blocked 5-fold, blocked 5-fold with a purge of h rows, and the last 20% of the in-set.
  2. Compute the true RMSE: the model fitted on the in-set, evaluated on the last 20% of the sample.
  3. Repeat for the target "mean $\log RV$ of the next 22 days".
  4. Interpretation: which scheme would you trust, and why are all of them below the true error?
- **Report**: ten RMSE values, one sentence
- Notebook: section B1

B1: solution (1/2) [Solved]





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B1: solution (2/2) [Solved]

target	random	blocked	purged	last 20%	true
next day	0.572	0.573	0.573	0.626	0.681
22-day mean	0.434	0.442	0.442	0.453	0.514

- Next day: random and blocked agree (HAR errors are nearly uncorrelated, the Bergmeir–Hyndman–Koo case); 22-day mean: random is lower (0.434 against 0.442): overlapping targets leak
- Interpretation: every scheme misses the 2020 volatility episode of the evaluation period: CV estimates the risk of the past regime; the last block is the most honest

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B2: a random forest with a time index [Proposed]

- **Question**: how much does a time-index feature inflate the optimism of random cross-validation?
- **Inputs**: DAX log RV (Oxford-Man); random forest (100 trees) on the 22 lags; target: mean log RV of the next 22 days; model: B1
- **Tasks**
  1. Compute the four CV estimates and the true RMSE with the lags only.
  2. Add the row number as a feature and repeat.
  3. Interpretation: which scheme is fooled by the time index, and which is not?
- **Report**: ten RMSE values, one sentence
- Notebook: section B2

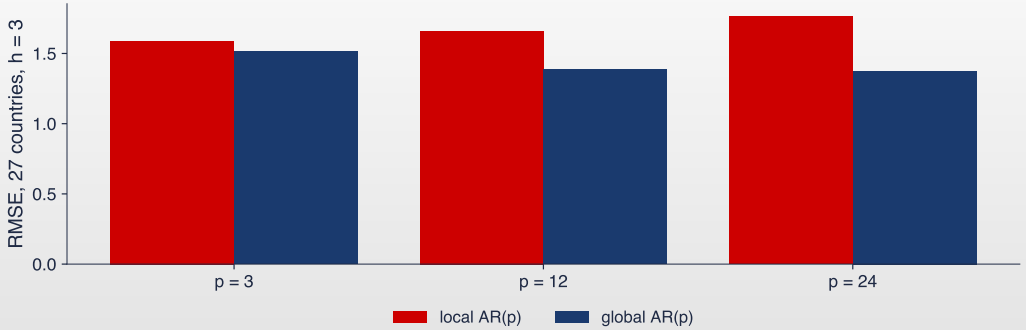
Back Exercise list

B3: local and global AR for EU inflation [Solved]

- **Question**: does one AR model for all 27 countries forecast annual inflation three months ahead better than 27 separate models?
- **Inputs**: Eurostat HICP annual rates, 27 EU countries, rolling 120-month windows, 137 origins from January 2015; direct forecasts $h = 3$
- **Tasks**
  1. Fit local AR(p) (one OLS per country) and global AR(p) (one pooled OLS) for $p = 3, 12, 24$.
  2. Compute the RMSE over the 27 countries and for Romania.
  3. Compute the DM statistic of global against local on the cross-country average loss.
  4. Interpretation: why does the gap widen with p?
- **Report**: twelve RMSE values, three DM statistics, one sentence
- Notebook: section B3

Back Exercise list

B3: solution (1/2) [Solved]



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B3: solution (2/2) [Solved]

p	local	global	DM (27)	p-value	local RO	global RO
3	1.589	1.514	-1.56	0.121	1.463	1.498
12	1.660	1.386	-2.29	0.024	1.327	1.373
24	1.766	1.376	-3.11	0.002	1.395	1.372

- Interpretation: local AR deteriorates with p (about 100 rows for $p+1$ coefficients), global AR improves (2700 rows); at $p = 24$ the global gain is significant ($p = 0.002$); for Romania alone the difference is not significant

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B4: the panel lasso for Romania [Proposed]

- **Question**: do the inflation rates of the other 26 EU countries help forecast Romanian inflation three months ahead?
- **Inputs**: 3 own lags (not penalised) and 3 lags of the 26 other countries; rolling 120-month windows, refit every 3 months; penalty by blocked 5-fold CV with a purge of 3 months; model: B3
- **Tasks**
  1. Forecast with AR(3), ridge, lasso, adaptive lasso, elastic net, post-lasso OLS and the global AR(12).
  2. Compute the RMSE and the DM test of each model against AR(3).
  3. Report the median number of selected predictors and the three most selected countries.
  4. Interpretation: is the high-dimensional panel worth it here?
- **Report**: seven RMSE values, six DM p-values, the selection summary, one sentence
- Notebook: section B4

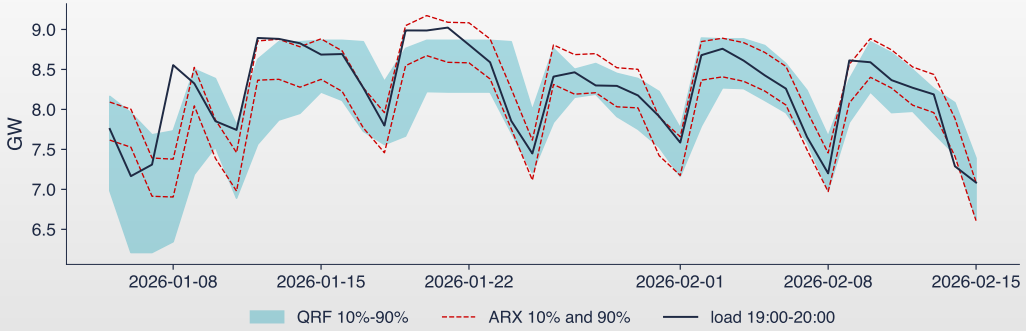
Back Exercise list

B5: QRF and ARX for the evening peak [Solved]

- **Question**: does a quantile regression forest give better day-ahead quantiles than the expert ARX for the load at 19:00–20:00?
- **Inputs**: Romanian hourly load, hour 19:00–20:00; 638 forecast days from January 2025; QRF (200 trees, leaves of at least 5) refitted monthly on all earlier data; ARX with empirical error quantiles (Chapter 1)
- **Tasks**
  1. Compute the deciles of both methods for each day.
  2. Compute the average pinball loss, the 80% coverage, the mean width of the 80% interval and the MAE of the point forecast.
  3. Test the difference of the daily pinball losses with DM.
  4. Interpretation: which bands would a grid operator prefer?
- **Report**: eight numbers, one DM statistic, one sentence
- Notebook: section B5

Back Exercise list

B5: solution (1/2) [Solved]



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B5: solution (2/2) [Solved]

	pinball (MW)	coverage 80%	width (MW)	MAE (MW)
QRF	78.5	86.7%	784	195
ARX	77.6	77.6%	530	187

- DM on the pinball difference: 0.40 ($p = 0.69$): no significant difference
- Interpretation: QRF bands are wider and over-cover (86.7%), ARX bands are sharper and slightly under-cover; equal scores, so the choice depends on the operator's cost of a shortfall

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B6: monotone boosting for the evening peak [Proposed]

- **Question**: does forcing the forecast to increase with the lagged loads cost accuracy at the evening peak?
- **Inputs**: the same rows as B5; histogram gradient boosting (300 iterations, 15 leaves), with and without increasing constraints on the three load lags, refitted monthly; model: B5
- **Tasks**
  1. Compute the MAE of both models and the DM test of their absolute errors.
  2. Count the decreasing steps of the partial dependence on the load of day d in the first fit.
  3. Interpretation: is the constraint worth its cost here?
- **Report**: two MAE values, one DM statistic, two counts, one sentence
- Notebook: section B6

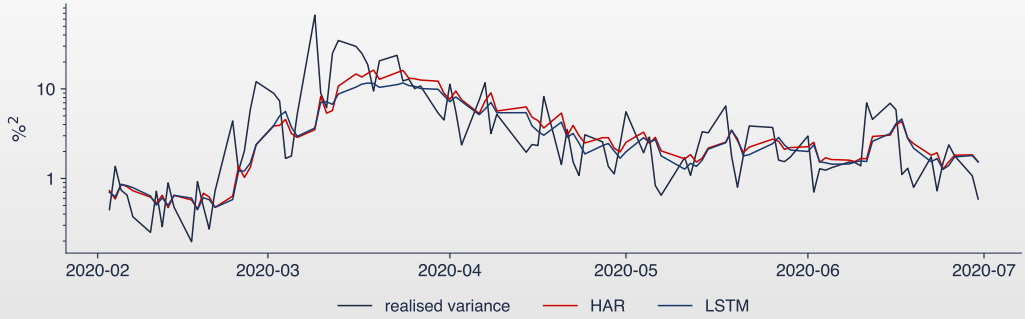
Back Exercise list

B7: LSTM against HAR for the FTSE 100 [Solved]

- **Question**: does an LSTM on the last 22 days forecast next-day realised variance better than HAR?
- **Inputs**: FTSE 100 log RV; expanding window from 2500 days, re-estimation every 500 days; forecasts 8 December 2009 – 25 February 2022 ($T = 3\,085$); LSTM with 16 units, average of three seeds
- **Tasks**
  1. Forecast $RV_{t+\mathbf{1}}$ with HAR and the LSTM, with the lognormal correction.
  2. Compute the average QLIKE of both and their ratio.
  3. Compute the DM statistic.
  4. Interpretation: would you replace HAR?
- **Report**: two losses, one ratio, one DM statistic, one sentence
- Notebook: section B7

Back Exercise list

B7: solution [Solved]



- QLIKE: HAR 0.2486, LSTM 0.2474; ratio 0.995; DM -0.63 ($p = 0.53$)
- Interpretation: the LSTM is indistinguishable from HAR; it costs seeds, tuning and minutes of training for no measurable gain: keep HAR, or average the two

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B8: DLinear and N-BEATS on 100 M4 hourly series [Proposed]

- **Question**: does a global N-BEATS still beat the benchmarks when it is trained on only 100 series?
- **Inputs**: the first 100 M4 hourly series, horizon 48, input 336 hours, one seed; model: B7
- **Tasks**
  1. Forecast with Naive 2, the seasonal naive, global DLinear and global N-BEATS.
  2. Compute sMAPE, MASE and OWA.
  3. Interpretation: compare with the lecture, where N-BEATS used all 414 series and three seeds.
- **Report**: twelve numbers, one sentence
- Notebook: section B8

Back Exercise list

C1: deep learners against HAR for crypto-assets [Proposed]

- **Question**: is crypto volatility, with its 24/7 trading and frequent regime changes, a place where learners beat HAR?
- **Inputs**: Bitcoin and Ether daily 5-minute RV from Binance one-minute prices, 2018–2026; expanding window from 1000 days, refit every 250 days; model: B7
- **Tasks**
  1. Write a five-line pre-registration: hypothesis, learners, horizons, loss and test, what would change your mind.
  2. Forecast with HAR, an MLP and boosting at $h = 1$ and $h = 5$; compute QLIKE ratios and DM tests against HAR.
  3. Interpretation: what do the data support, and how would you extend the study?
- **Report**: a five-line plan, eight ratios with p-values, a project plan
- Notebook: section C1

Back Exercise list

C2: audit an AI answer [Proposed]

- A student asked an AI assistant to summarise machine learning for time series. The answer:
- (a) Random K-fold cross-validation is never valid for time series.
- (b) A TCN with kernel 3 and dilations 1, 2, 4, 8 (two convolutions per block) sees the last 31 hours.
- (c) With enough trees, bagging drives the variance of the forecast to zero.
- (d) LSTMs never suffer from vanishing gradients, thanks to their gates.
- (e) Zeng et al. (2023) proved that every Transformer is worse than a linear model.
- (f) The patch with the largest attention weight is the one the forecast depends on most.
- (g) Shapley values of all features sum to the prediction minus the average prediction over the background.
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
  - 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- Validate with blocks and purges when targets overlap or features carry time; random K-fold only for well-specified autoregressions
- Pooling series (global models) buys memory; selection on short panels is unstable
- Ensembles: the variance floor is $\rho\sigma^2$; boosting fits gradients; constraints must be right everywhere
- Gates and attention have simple algebra: forget gates must be open, attention ignores order
- Strong baselines (HAR, expert ARX, seasonal naive) are hard to beat; test every claim

After the seminar

- Lecture 12 develops each topic of today: learning theory for dependent data, high-dimensional inference, tree ensembles, recurrent and attention models, N-BEATS, DeepAR and honest benchmarking
- C1 can grow into a team project: learners against HAR for crypto-assets, pre-registered
- Reading: Bergmeir, Hyndman and Koo (2018); Montero-Manso and Hyndman (2021); Zeng et al. (2023); Oreshkin et al. (2020)
- **The seminar is for practice and is not graded**

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